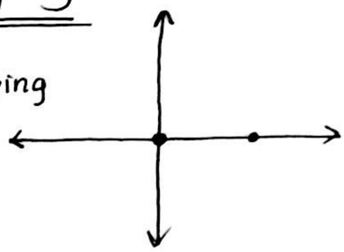
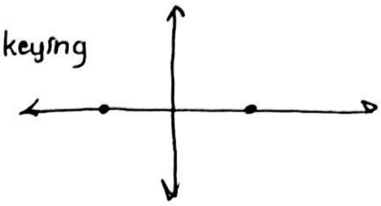
Binh Phan

EE558 HW #5

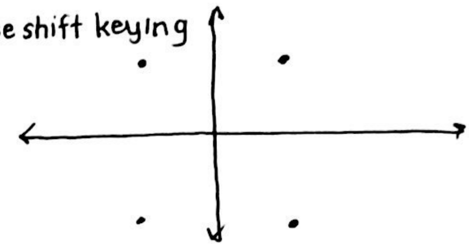
1. a. ASK - amplitude shift keying



b. BPSK - binary phase shift keying



c. QPSK - quadrature phase shift keying



b. QPSK $\rightarrow$ 2 bits/symbol time

2. $BER \approx 0.5\,\mathrm{erfc}(Q/\sqrt{2})$

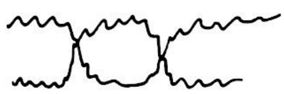
$BER = 10^{-3} \rightarrow Q \approx 3$

$BER = 10^{-9} \rightarrow Q \approx 6$

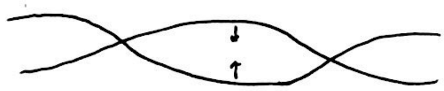
3. $Q = \dfrac{(V_1 - V_0)}{(\sigma_1 + \sigma_0)}$, $\sigma_1 = \sigma_0 = 50nV$, $V_0 = 0$

$BER = 10^{-9} \rightarrow Q \approx 6 = \dfrac{V_1}{100nV}$, $\underline{V_1 = 600nV}$

4. Increase LPF cutoff frequency $\rightarrow$ Eye diagram gets noisier



Decrease LPF cutoff frequency $\rightarrow$ Eye starts to close



5. N_p is the # of photons/bit

For $BER = 10^{-9}$, ~~N_p:~~ ~~PSK (heterodyne) → BER = ½erfc~~

ASK (heterodyne) will have the highest N_p $\rightarrow$ highest sensitivity